INCH-POUND

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DETAIL SPECIFICATION

INITIATOR, PROPELLANT ACTUATED, JAU-2/A25 ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the JAU-2/A25 Propellant Actuated Initiator Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4 or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-A-2550	Ammunition, General Specification For
MIL-STD-1168	Ammunition Lot Numbering and Ammunition Data Card
MS28741	Hose Assembly, Detachable End Fitting, Medium Pressure

(Copies of these documents are available online at http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

Comments, suggestions, or questions on this document should be addressed to Commander, Indian Head Division, Naval Surface Warfare Center, Technical Information Branch (Code 4230), 101 Strauss Avenue, Indian Head, Maryland 20640-5035, or emailed to ~~amanda.penn@navy.mil~~. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at ~~http://assist.daps.dla.mil~~.

AMSC N/A FSC 1377

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DRAWINGS

838AS163	Test Fixture, 6.5 Cubic Inch Closed Bomb
9280894	Box, Fiberboard (Inner), Packing, Ammunition, For Initiator, (Gas-Subminiature)
9280895	Box, Packing, Ammunition, For Initiator, (Gas-Subminiature)
9280896	Box, Fiberboard (Outer), Alternative, Packing Ammunition, for Initiator, (Gas Subminiature)
DL11731494	Initiator, Propellant Actuated, JAU-2/A25 Assembly

(Application for copies should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Technical Information Branch (Code 4230), Indian Head, MD 20640-5035.)

U.S. Air Force

SPI-F00-138-3770 Initiators

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E1742 Standard Practice for Radiographic Examination

(Copies of this document are available online at http://www.astm.org/ or from the American Society for Testing and Materials Customer Service, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

AMERICAN NATIONAL STANDARD INSTITUTE/AMERICAN SOCIETY FOR QUALITY

ANSI/ASQ Zl.4 Sampling Procedures and Tables for Inspection by Attributes

ANSI/ASQ A8402 Quality Management and Quality Assurance – Vocabulary

(Copies of these documents are available online at http://webstore.ansi.org/ansidocstore/default.asp, or from the American Society for Quality, PO Box 3005, 611 E. Wisconsin Ave., Milwaukee, WI 53201-4606.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Conformance to documents. The JAU-2/A25 initiator covered by this specification shall be manufactured in accordance with DL11731494.

3.3 Materials. All materials used in the manufacture of the initiators shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.4 Components.

3.4.1 Primary components. For the purpose of this specification, the PVU-1/A Ignition Device (Drawing 851AS110) is defined as a primary component. Only primary components from a single lot shall be used in a production lot of initiators; however, one primary component production lot may be used in more than one initiator production lot.

3.4.2 Ignition device. The ignition device shall conform to the applicable drawing and reference specification. Prior to assembly into an initiator, the ignition device shall be visually examined in accordance with 4.5.2.5 and radiographically examined for defects as specified in 4.6.6.1. The ignition device shall be manufactured within 2 years of assembly into the ignition device retainer (11738408) or recertified within 12 months of assembly into the ignition device retainer.

3.4.3 Ignition device sensitivity. The PVU-1/A Ignition Device, when assembled and crimped into an ignition device retainer (Drawing 11738408), shall fire when drop tested at an energy level of 25.48 inch ounces, maximum (see 4.5.3.6).

3.4.4 Leak test. Each retainer and ignition device assembly, Drawing 11738409, and the M104 propellant actuated initiator, P/N 10551446, shall withstand 14 psi internal-external pressure differential with a maximum permissible leak rate of 1×10^{-4} cubic centimeters per second of air (see 4.5.3.5).

3.5 Breakaway torque. Each threaded joint of the assembled initiator shall withstand a breakaway torque application of 400 inch-pounds without loosening or deformation (see 4.5.3.4).

3.6 Residual magnetism. The initiator shall not deflect the indicator of the compass more than five degrees in either direction (see 4.5.3.3).

3.7 Radiographic examination. Each ignition device assembly, retainer and ignition device assembly, and initiator assembly shall be radiographically examined for defects (see 4.6.6).

3.8 Functional.

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3.8.1 Ballistics. Initiators loaded with accepted pyrotechnic components (II-A pellets and PVU-1/A ignition device) and temperature conditioned shall produce pressure values specified in table I. Pressure values shall be recorded in a 6.5 cubic inch closed bomb, Drawing 838AS163.

3.8.1.1 Initiation. The initiator shall be fired by a source delivering an output pressure of 700 +25, -0 pounds per square inch gage (psig) at 70 ± 5°F.

3.8.2 Misfires. There shall be no misfires.

3.8.3 Mechanical failures. The initiator shall function without mechanical failure (see 6.5.2).

TABLE I. Pressure values.

Temperature (°F)	Pressure (psig)	Ignition delay (see 6.5.1) milliseconds, max
	Minimum	
-60 to -70	300	10
65 to 75	300	10
195 to 205	300	10

3.9 Shear pin strength. The shear pin, after application of protective finish, shall have a single shear when a load of 41 ± 5 pounds is applied at the steady rate of 0.125 inch per minute (see 4.5.3.1).

3.10 Protective finish. Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of configuration (see 4.5.3.2).

3.11 Workmanship. Initiators shall be free of cracks, splits, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.11.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.11.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.11.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Conformance inspection (see 4.5).

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4.2 Quality assurance terms and definitions. Reference shall be made to ANSI/ASQ A8402 to define the quality assurance terms used.

4.3 Inspections. Inspections shall be in accordance with MIL-A-2550 and ASTM E1742.

4.3.1 Inspection conditions. Unless otherwise specified in the contract or purchase order (see 6.2), inspection conditions shall be as follows:

a. Temperature: Room ambient 20° to 25° C (68° to 77° F)
b. Altitude: Sea level
c. Humidity: 50 percent relative, maximum
d. Atmospheric: 14.7 psia

4.4 First article inspection. At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 21 assembled initiators for ballistic tests, a quantity of ignition devices in accordance with table II assembled and crimped into the retainer, five metal parts assemblies, fifty shear pins, and nine sets of unassembled parts for visual and dimensional examination and tests, and one set of unassembled components without plating or final protective finish. The sample shall be manufactured using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging, shall be the same as used in regular production and shall be obtained from the same source of supply.

4.4.1 Examination, tests, and facility. After provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or other such facility specified in the contract (see 6.2).

4.4.2 Sample size for primer sensitivity. The number of ignition devices, assembled and crimped into the retainer, submitted for ignition device sensitivity tests shall be in accordance with table II.

TABLE II. Ignition device sensitivity.

Lot Size	Sample Size
3 to 15	10
16 to 25	10
26 to 40	10
41 to 65	10
66 to 110	10
111 to 180	15
181 to 300	25
301 to 500	30
501 to 800	35
801 to 1300	40
Over 1300	50

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4.4.3 Temperature conditioning for ballistic test. The sample shall be temperature conditioned and tested to determine compliance with 3.8.1. The sample for firing shall be conditioned to the temperatures specified in table III.

TABLE III. Temperature conditioning for first article sample.

Temperature (°F)	Not vibrated	Vibrated
-60 to -70	3	4
65 to 75	3	4
195 to 205	3	4

4.4.4 First article failure. Failure of the sample to comply with the requirements of the drawings and specifications shall result in first article sample disapproval.

4.5 Quality conformance inspection.

4.5.1 Lot.

4.5.1.1 Submission of product. The product shall be submitted in accordance with ANSI/ASQ Z1.4.

4.5.1.2 Lot size. The production lot size shall be as specified in the contract (see 6.2).

4.5.2 Examination. One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed in accordance with the classification of defects. Four power (4x) magnification may be used for inspection of such defects as burr and foreign matter. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gauged to determine acceptability. All nonconforming material shall be rejected.

4.5.2.1 Pin, Firing, Drawing 11738410, covering a detail of Drawing 11731492.

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
101	True position of O-ring groove with body	Gage
102	Concentricity of striker with body	Gage
103	True position of shear pin hole with body	Gage
104	Diameter of O-ring groove, min	Gage
105	Diameter of body, min	Gage
106	Diameter of striker	Gage
107	Length of striker, min	Gage
108	Diameter of shear pin hole, max	Gage
109	Depth of shear pin hole, max	Gage
110	Width of O-ring groove	Gage

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111	Location of shear pin hole	Gage
112	Permissible variation of O-ring groove	Gage
113	Spherical radius of tip	Gage
114	Metal defective	Visual
115	Protective finish inadequate or defective	Visual
116	Burr or sharp corner, except where permitted	Visual
Minor		
201	Diameter of O-ring groove, max	Gage
202	Diameter of body, max	Gage
203	Diameter of shear pin hole, min	Gage
204	Depth of shear pin hole, min	Gage
205	Width of slot	Gage
206	Depth of slot	Gage
207	Total length	Gage
208	Length of body to O-ring groove	Gage
209	location of marking	Visual
210	Marking missing, incorrect, or illegible	Visual
211	Surface finish improper, except as otherwise classified	Visual

4.5.2.2 Housing, Drawing 11731622, covering a detail of Drawing 11731492.

Characteristics	Defect	Method of Inspection
Critical		
1	Metal defective	Visual
2	Surface finish of cavity	Gage
Major		
101	Pitch diameter of external thread, min	Gage
102	Major diameter of external thread, min	Gage
103	Pitch diameter of internal thread, max	Gage
104	Minor diameter of internal thread, max	Gage
105	True position of pitch diameter of external thread with end flange diameter	Gage
106	Concentricity of pitch diameter of internal thread on large end with center cavity	Gage
107	True position of pitch diameter of internal thread with center cavity	Gage
108	True position of striker hole with center cavity	Gage
109	True position of shear pin hole with center cavity	Gage
110	True position of flats with internal thread	Gage
111	Perpendicularity of striker face with center cavity	Gage
112	Perpendicularity of striker face with internal thread on large end	Gage
113	Diameter of shear pin hole	Gage
114	Diameter of shear pin counterbore	Gage
115	Diameter of striker hole	Gage
116	Diameter of bore, large end	Gage
117	Depth of bore, large end	Gage

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118	Diameter of bore, small end	Gage
119	Depth of bore, small end	Gage
120	Depth of shear pin counterbore hole	Gage
121	Length of shear pin hole	Gage
122	Length of internal thread on large end	Gage
123	Length of internal thread on small end, min	Gage
124	Length of internal thread on small end, max	Gage
125	Width of thread undercut	Gage
126	Location of shear pin hole	Gage
127	Diameter of center cavity	Gage
128	Width of striker plate	Gage
129	Diameter of counterbore, small end	Gage
130	Chamfer at mouth of small end	Gage
131	Thread lead angle	Gage
132	Protective finish inadequate or defective	Visual
133	Burr or sharp corner	Visual
134	Foreign matter	Visual
Minor		
201	Pitch diameter of external thread, max	Gage
202	Pitch diameter of internal threads, min	Gage
203	Width across flats	Gage
204	Diameter of flange	Gage
205	Width of flange	Gage
206	Depth of counterbore	Gage
207	Location of external thread from face	Gage
208	Location of shoulder from face	Gage
209	Width of shoulder	Gage
210	Diameter of shoulder	Gage
211	Total length	Gage
212	Diameter of large end	Gage
213	Angle at bottom of bore, small end	Gage
214	Surface finish improper, except as otherwise classified	Visual

4.5.2.3 Pin, Shear, Drawing 11731626, covering a detail of Drawing 11731492.

Characteristics	Defect	Method of Inspection
Critical		
1	Metal defective	Visual
Major		
101	Diameter, min	Gage
102	Total length, min	Gage
103	Surface finish improper	Visual
104	Protective finish inadequate or defective	Visual
105	Burr	Visual
106	Foreign matter	Visual

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Minor		
201	Diameter, max	Gage
202	Total length	Gage

4.5.2.4 Bracket, Mounting, Drawing 11731741, covering a detail of Drawing 11731494.

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
101	Pitch diameter of internal thread, max	Gage
102	Minor diameter of internal thread, max	Gage
103	True position of mounting holes with pitch diameter of internal thread	Gage
104	Length of effective thread, min	Gage
105	Length of thread, max	Gage
106	Metal defective	Visual
107	Protective finish inadequate or defective	Visual
108	Burr	Visual
109	Foreign matter	Visual
Minor		
201	Diameter of mounting holes	Gage
202	Length across thread of bracket	Gage
203	Length across end of bracket (front)	Gage
204	Total length	Gage
205	Location of mounting holes	Gage
206	Location of large hole	Gage
207	Height of shoulder	Gage
208	Total height including boss	Gage
209	Diameter of bore	Gage
210	Radii incorrect	Gage
211	Surface finish improper	Visual

4.5.2.5 Ignition Device (Percussion), Drawing 851AS110, covering a detail of Drawing 11738409.

Characteristics	Defect	Method of Inspection
Critical		
1	Improper assembly	Visual/X-ray
2	Any part missing	Visual/X-ray
3	Pellet broken, cracked, or dislodged	Visual
4	Metal defective	Visual
5	Sealing foil discolored	Visual
6	Mixed ignition devices	Visual

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Major		
101	Burr	Visual
102	Foreign matter	Visual
Minor		
None defined		

4.5.2.6 Ignition Device, Retainer, Drawing 11738408, covering a detail of Drawing 11738409.

Characteristics	Defect	Method of Inspection
Critical		
1	Diameter of ignition device pocket	Gage
2	Depth of ignition device pocket	Gage
3	Metal defective	Visual
Major		
101	Thickness of wall between ignition device pocket and relief cavity	Gage
102	Outside diameter	Gage
103	True position of relief cavity with ignition device cavity	Gage
104	True position of body with ignition device cavity	Gage
105	Protective finish inadequate or defective	Visual
106	Surface finish of ignition device pocket improper	Visual
107	Burr or sharp corner	Visual
108	Foreign matter	Visual
Minor		
201	Diameter across chamfer	Gage
202	Diameter of relief cavity	Gage
203	Total length	Gage
204	Radius or chamfer missing or incorrect	Visual
205	Surface finish improper, except as otherwise classified	Visual

4.5.2.7 Retainer and Ignition Device Assembly, Drawing 11738409, covering a detail of Drawing 11731494.

Characteristics	Defect	Method of Inspection
Critical		
1	Ignition device not properly seated	Gage
Major		
101	Ignition device not properly sealed and crimped	Visual
Minor		
None defined		

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4.5.2.8 Initiator, M104 Assembly, Drawing 10551446, covering a detail of Drawing 11731494.

Characteristics	Defect	Method of Inspection
Critical		
1	Mixed ammunition	Visual
2	Metal defective	Visual
3	Vent hole obstructed	Visual
4	Marking missing, incorrect, or illegible	Visual
Major		
101	Thread damaged	Visual
102	Burr	Visual
103	Foreign matter	Visual
Minor		
None defined		

4.5.2.9 Firing Mechanism Assembly, Drawing 11731492, covering a detail of Drawing 11731494.

Characteristics	Defect	Method of Inspection
Critical		
1	Firing pin protrusion	Gage
2	Shear pin not engaged in firing pin	Gage
3	Striker hole in initiator housing obstructed	Visual
Major		
101	Burr	Visual
102	Foreign matter	Visual
Minor		
None defined		

4.5.2.10 Assembly, Drawing 11731494.

Characteristics	Defect	Method of Inspection
Critical		
1	Any part missing	X-ray/Visual
2	Improper assembly	X-ray/Visual
3	Total length, max	Gage
4	Marking missing, incorrect, or illegible	Visual
Major		
101	Burr	Visual
102	Foreign matter	Visual

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Minor		
201	Thread sealing compound missing	Visual
202	Protective cap or plug missing	Visual
203	Protective finish inadequate or defective	Visual

4.5.2.11 Box, Fiberboard (Inner), Ammunition, Drawing 9280894.

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
None defined		
Minor		
201	Contents loose	Manual
202	Box damaged	Visual
203	Sealing strip broken or missing	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Cut, damaged, or improperly sealed bag	Visual

4.5.2.12 Box, Ammunition, Packing, Drawing 9280895.

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
None defined		
Minor		
201	Contents loose	Manual
202	Strapping missing, loose, broken, or improperly engaged	Visual/Manual
203	Contents of box exposed	Visual
204	Box damaged	Visual
205	Marking missing, incorrect, or illegible	Visual
206	Number of units incorrect	Visual

4.5.2.13 Box, Fiberboard (Outer), Alternative, Ammunition, Packing, Drawing 9280896.

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
None defined		

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Minor		
201	Contents loose	Manual/Visual
202	Carton broken or damaged	Visual
203	Number of units contained incorrect	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Tape incomplete	Visual

4.5.3 Inspection testing. The following tests shall be performed in accordance with 4.6.

4.5.3.1 Shear pin strength. Fifty pins from each lot of shear pins tested by the contractor shall be tested for single shear strength. Failure of any pin to comply with 3.9 shall be cause for rejection of the lot.

4.5.3.2 Protective finish. Unless otherwise specified, sampling plans and procedures in determination of the acceptability of coated parts and articles submitted by a supplier shall be in accordance with 3.10.

4.5.3.3 Residual magnetism. Each assembly shall be subjected to the test for excessive residual magnetism. Failure of any assembly to comply with 3.6 shall be cause for rejection of the initiator.

4.5.3.4 Breakaway torque. Each initiator shall be subjected to the breakaway torque test. Failure of any assembly to comply with 3.5 shall be cause for rejection of the assembly.

4.5.3.5 Leak test. Each M104 Initiator and each retainer and ignition device assembly shall be subjected to a leak test. Failure of any M104 Initiator or retainer and ignition device assembly to comply with 3.4.4 shall be cause for rejection of the M104 Initiators or retainer and ignition device assembly. This test must be done by the contractor before final assembly.

4.5.3.6 Ignition device sensitivity. A sample of ignition devices, selected in accordance with table II from the initial production lot and each subsequent lot, shall be assembled and crimped into the retainer and tested for sensitivity.

4.5.3.6.1 All fire. One half the ignition device sample size, crimped into the retainer, shall be tested at the 25.48 inch-ounce energy level in accordance with 3.4.3. Any ignition device failing to comply with 3.4.3 shall be cause for rejection.

4.5.3.7 Functional tests. The following tests shall be performed by the Government.

4.5.3.7.1 Ballistics. Twenty-one units shall be tested to determine compliance with 3.8. The units shall be temperature conditioned as provided in table IV. Failure of any unit to comply with 3.8 shall be cause for rejection of the lot.

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TABLE IV. Temperature conditioning for quality conformance sample.

Temperature (°F)	Sample Size
-60 to -70	7
65 to 75	7
195 to 205	7

4.5.3.7.2 Retest. There shall be no retests.

4.5.3.7.3 Test failure. If a test failure is attributable to an assignable cause, excluding the test initiator, the original test results shall be discarded and that test reconducted.

4.6 Test methods and procedures.

4.6.1 Shear pin strength. After application of protective finish, the pin shall be subjected to the single shear strength test to determine compliance with 3.9. The test shall be performed in the fixture in conjunction with a tensile testing machine.

4.6.2 Protective finish. Test procedures for protective finish shall be in accordance with the specification referenced on the component drawing.

4.6.3 Breakaway torque. A torque wrench, set at the torque requirement specified in 3.5, shall be used in conjunction with the fixture to determine the adequacy of the joint. Torques shall be applied in three places: (1) between the female body and male body; (2) between the male or female body and the firing mechanism assembly; and (3) between the mounting bracket and the firing mechanism assembly.

4.6.4 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end at the same height as the compass, in the north-south horizontal position (cap end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis, and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the assembly for four full passes.

4.6.5 Leak test. The method and procedure for performing the leak test shall be as approved by the Government.

4.6.6 Radiographic examination. All ignition device assemblies, retainer and ignition device assemblies, and initiator assemblies shall be radiographically examined for defects.

4.6.6.1 Ignition device assembly, Drawing 851AS110. Prior to assembly into the cartridge, the ignition device shall be radiographically examined in accordance with ASTM E1742. Each ignition device shall be X-rayed on boards or trays with the cup side perpendicular to the rays. The X-ray negatives shall be identified by item nomenclature and lot number. Ignition devices are to be left mounted to the boards or trays until after the X-rays have been reviewed and defects identified to facilitate removal of defective units. After screening, the X-rays with defectives identified shall be forwarded to the engineering agency

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designated by the procuring activity. Film interpretation of the radiographs shall verify the presence and correct positioning of the following:

a. Presence of ignition device charge
b. No material defects

4.6.6.2 Retainer and ignition device assembly, Drawing 11738409. All retainer and ignition device assemblies shall be radiographically examined in accordance with ASTM E1742, with the exception that a retainer and ignition device assembly shall be used for the penetrameter. Any observable imperfections in components or assembly shall be cause for rejection. The retainer and ignition device assemblies shall be positioned for the most revealing exposure, with the long axis of each assembly perpendicular to the rays of the X-ray machine. All retainer and ignition device assemblies shall be identified with serial numbers prior to X-ray examination. The serial numbers shall be in consecutive numerical order beginning with the number (001) in each production lot. Any missing serial numbers are to be identified on the X-rays. The retainer and ignition device assemblies shall be arranged on boards or trays in consecutive order and each radiograph shall carry a permanent identification of the units displayed thereon. The radiographic identification shall include item nomenclature, the complete lot number assigned to that lot, and the span of serial numbers displayed. Radiographs of the entire production lot shall be submitted to the activity conducting the production lot acceptance tests. Defective units found during X-ray review are to be marked on the X-ray prior to shipment and those defective items shall be removed from the production lot. Any assembling of the retainer and ignition device assembly into initiators prior to notification by the contracting agency of X-ray acceptance for the retainer and ignition device assembly shall be at the contractor's risk. Film interpretation of the radiographs shall verify the presence and correct positioning of the following:

a. Ignition device charge present and not cracked
b. Anvil present, seated, and not cocked
c. No material defects

4.6.6.3 Initiator assembly, Drawing 11731494. All assembled initiators shall be radiographically examined in accordance with ASTM E1742, with the exception that an initiator assembly will be used for the penetrameter. Any observable imperfections in components or assembly shall be cause for rejection. The initiators shall be positioned for the most revealing exposure, with the long axis of each assembly perpendicular to the rays of the X-ray machine. All initiators shall be identified with serial numbers prior to X-ray examination. The serial numbers shall be in consecutive order beginning with the number 001 in each production lot. Any missing numbers are to be identified on the X-rays. The initiators shall be arranged on trays or boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the units displayed thereon. The radiographic plate identification shall include item nomenclature, the complete lot number, the span of serial numbers displayed, and the contract number. The radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance test. Defective initiators found during X-ray review are to be marked on the X-ray prior to shipment and those defective items shall be removed from the production lot. Film interpretation of the radiographs shall verify the presence and correct positioning of the following parts:

a. Ignition device retainer, P/N 11738408
b. Ignition device, P/N 851AS110
c. M104 Assembly, P/N 10551446
d. Propellant, MIL-P-46994
e. Housing, P/N 11731622
f. Shear Pin, P/N 11731626
g. Firing Pin, P/N 11738410

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h. O-ring, MS28775-009
i. Mounting Bracket, P/N 11731741

4.6.7 Ignition device sensitivity. Drop test a sample of ignition devices, assembled and crimped into the retainer, selected in accordance with table II. Use a steel ball 15/16 inches in diameter and weighing 1.94 ± 0.02 ounces at a drop height of 13 inches (measured from bottom of suspended ball to top of firing pin) in a primer sensitivity machine approved by the procuring agency.

4.6.8 Functional.

4.6.8.1 Ballistics. The test sample shall be rigidly supported in the fixture and fired by a source delivering an output of 700 +25, -0 psig at 70 ± 5°F. The sample shall be tested within 5 minutes after removal from the temperature conditioning chamber. When this time limit is exceeded, the sample units shall be reconditioned in accordance with 4.6.8.2. Required pressure readings shall be taken as specified in 3.8.1.

4.6.8.2 Temperature conditioning. Each assembled initiator shall be conditioned for not less than 4 hours nor greater than 24 hours. When reconditioning initiators at -60°F, all condensation shall be removed from the exterior surfaces before returning the initiator to the conditioning chamber.

4.7 Inspection of packaging. The sampling and inspection of the packaging, packing, and container marking shall be in accordance with section 5.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The initiator assembly is used in conjunction with other propellant actuated devices in Aircrew Emergency Escape Systems. Since the initiator was designed specifically for military use, there is no commercial application.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and revision letter of this specification

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2)

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c. Whether a first article sample is required (see 3.1, 4.4, and 6.4)

d. Test facility for first article tests (see 4.4)

e. Production test activity and lot size (see 4.1 and 4.5.1.2)

f. Directions for shipping radiographs of entire production lot along with ballistic sample to the activity conducting production lot acceptance tests (see 4.6.6)

g. Level of packing and marking (see 5.1 and 6.7)

h. That the safety precaution requirements of the ``Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of ``Ammunition and Explosives Ashore," OP 5 are applicable.

6.3 Hazard notice. The initiator described herein and certain of its components are flammable or explosive and present hazards in manufacture, handling, storage, and shipment. The contractor should recognize these properties and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling, and electrically induced incidents.

6.4 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items should be a first article sample or a sample selected from the first production items (see 3.1), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Definitions.

6.5.1 Ignition delay. Ignition delay time is the time measured between the firing pin striking the ignition device and the beginning of a continuous thrust rise on a thrust-time record.

6.5.2 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.6 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

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6.7 Contract packaging, packing and marking. The following requirements should be specified in the contract.

6.7.1 Packing and marking should be Level A or B, as specified by the contracting activity.

6.7.1.1 Level A. Level A should be in accordance with Drawings 9252247 and 9252248 or SPI F00-138-3770.

6.7.1.2 Level B. Level B should be in accordance with Drawings 9252247 and 9252249 or SPI F00-138-3770.

6.7.1.3 Lot numbers. Lot numbers should be in accordance with MIL-STD-1168.

6.8 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes. This revision was authorized by ECP number 045220I031.

Custodian:
Navy - OS
Air Force - 99
Review activity:
Air Force – 70

Preparing activity
Navy - OS
(Project 1377-2005-006)

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